



Temperature and Humidity sensor

Using DHT11 and Arduino

Dr: Tarek Elewa

Agenda

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02. Features

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Specification**

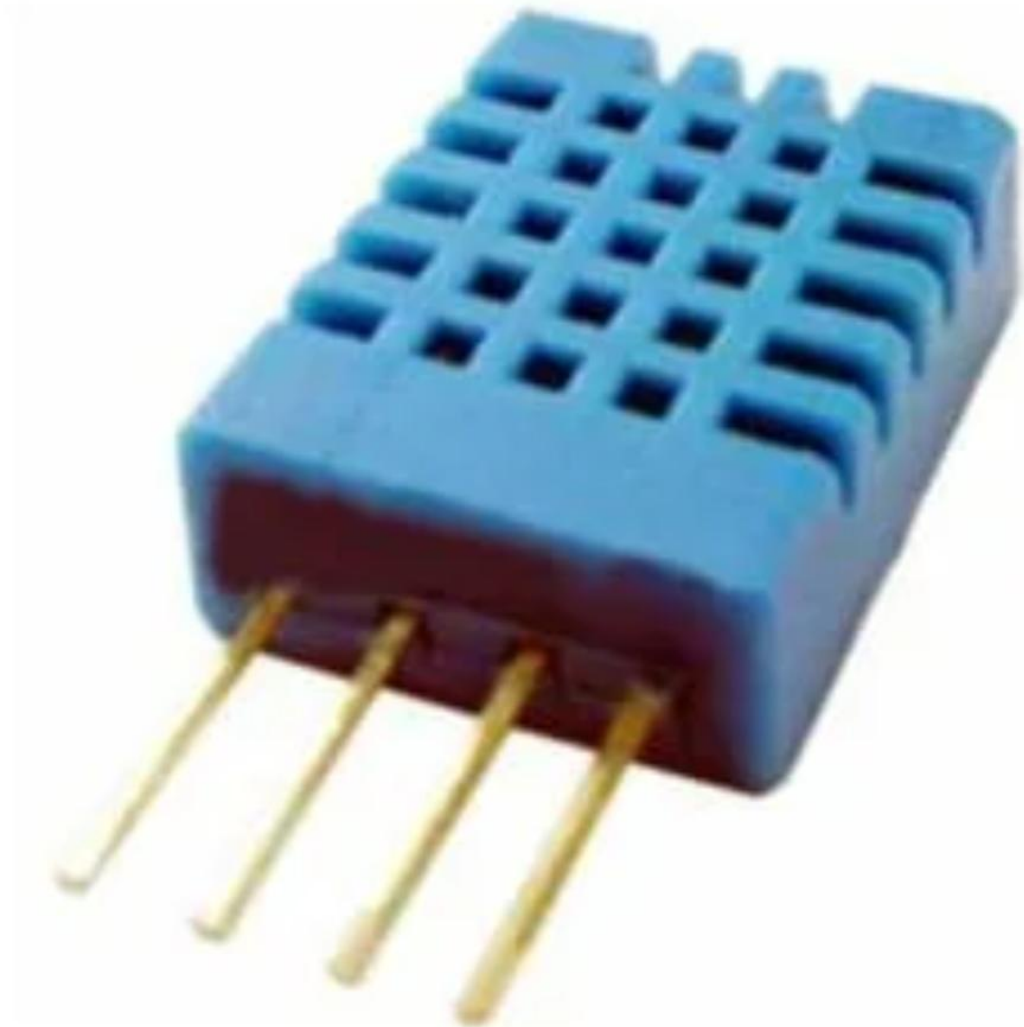
04. Operating

**05. Connecting
with Arduino**

Overview

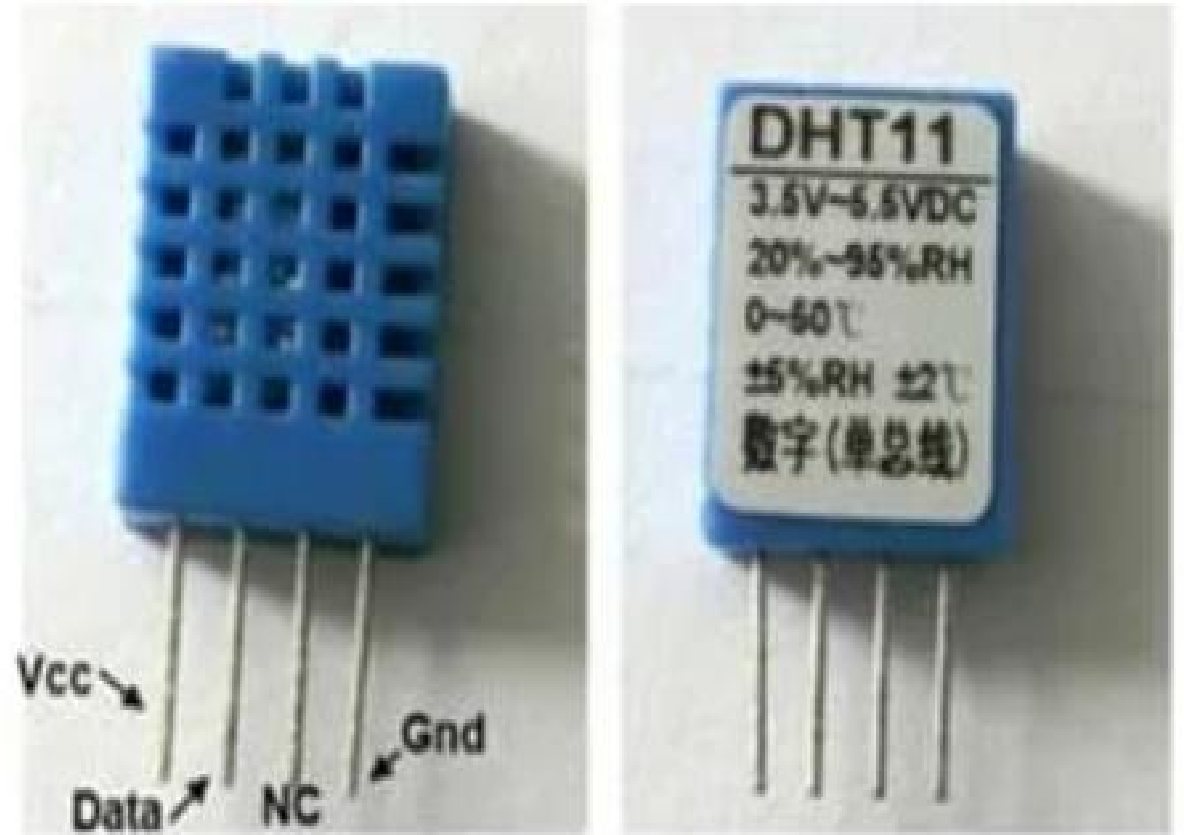
This sensor include a resistive type humidity measurement component and a NTC temperature measurement component ,and connect to a high performance 8-bit microcontroller ,offering excellent quality , fast response, anti-interface ability and cost effectiveness.

It's of small size, low power consumption and up-to 20 meter signal transmission makes it a best choice for various application



Features

- Humidity Range: 20-90% RH
- Humidity Accuracy: $\pm 5\%$ RH
- Temperature Range: 0-50 °C
- Temperature Accuracy: $\pm 2\%$ °C
- Operating Voltage: 3V to 5.5V



Technical Specification

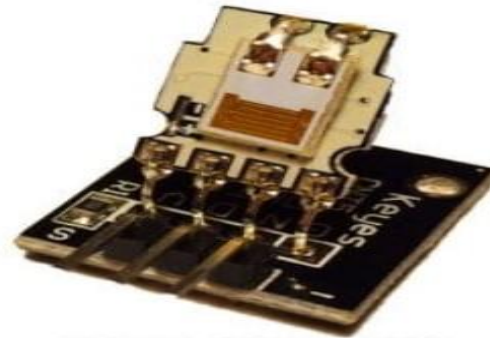
MODEL	DHT11
Power Supply	3-5.5V DC
Output Signal	Digital Signal via single-bus
Sensing element	Polymer resistor
Measuring range	Humidity 20-90%RH Temperature 0-50 Celsius
Accuracy	Humidity $\pm 4\%RH$ (Max $\pm 5\%RH$); temperature ± 2.0 Celsius
Resolution or sensitivity	Humidity 1%RH;
Repeatability	Humidity $\pm 1\%RH$
Humidity Hysteresis	$\pm 1\%RH$
Long term Stability	$\pm 0.5\%RH$ /year
Sensing period	Average: 2s
Dimensions	Size 12*15.5*5.5mm

Operation of Measuring

- The DHT11 detects water vapor by measuring the electrical resistance between two electrodes. The humidity sensing component is a moisture holding substrate with electrodes applied to the surface. When water vapor is absorbed by the substrate, ions are released by the substrate which increases the conductivity between the electrodes. The change in resistance between the two electrodes is proportional to the relative humidity. Higher relative humidity decreases the resistance between the electrodes, while lower relative humidity increases the resistance between the electrodes.
- The DHT11 measures temperature with a surface mounted NTC temperature sensor (thermistor) built into the unit.

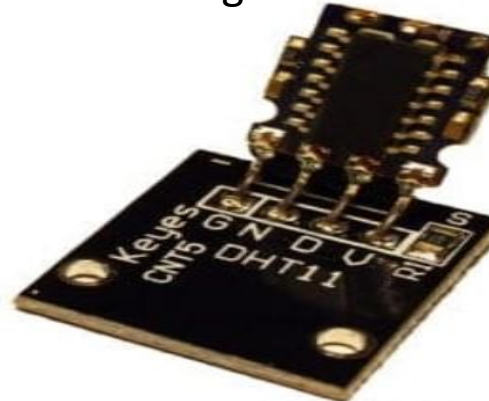
Operation of Measuring

- With the plastic housing removed, you can see the electrodes applied to the substrate:

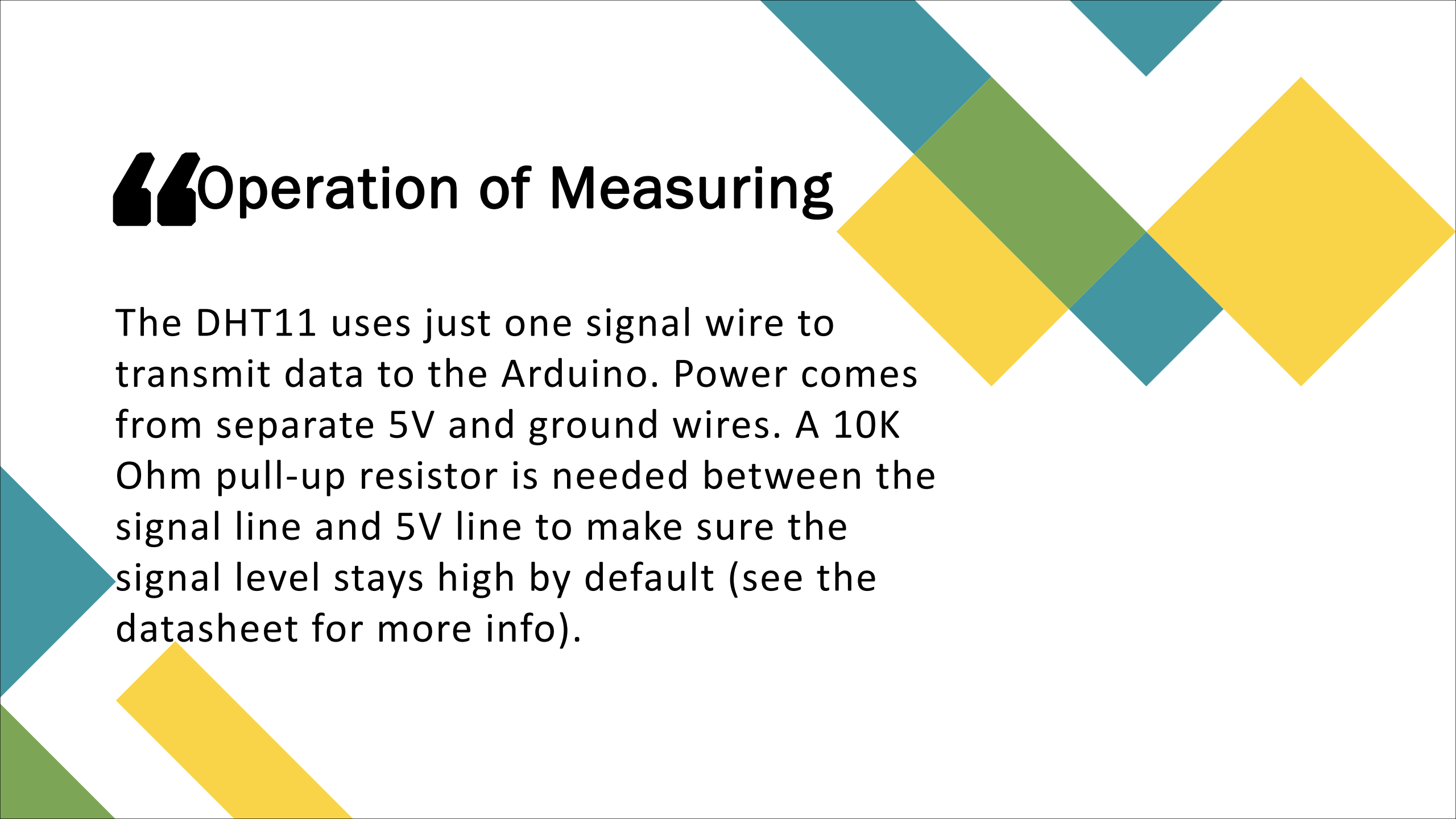


Front view with
cover removed

An IC mounted on the back of the unit converts the resistance measurement to relative humidity. It also stores the calibration coefficients, and controls the data signal transmission between the DHT11 and the Arduino:



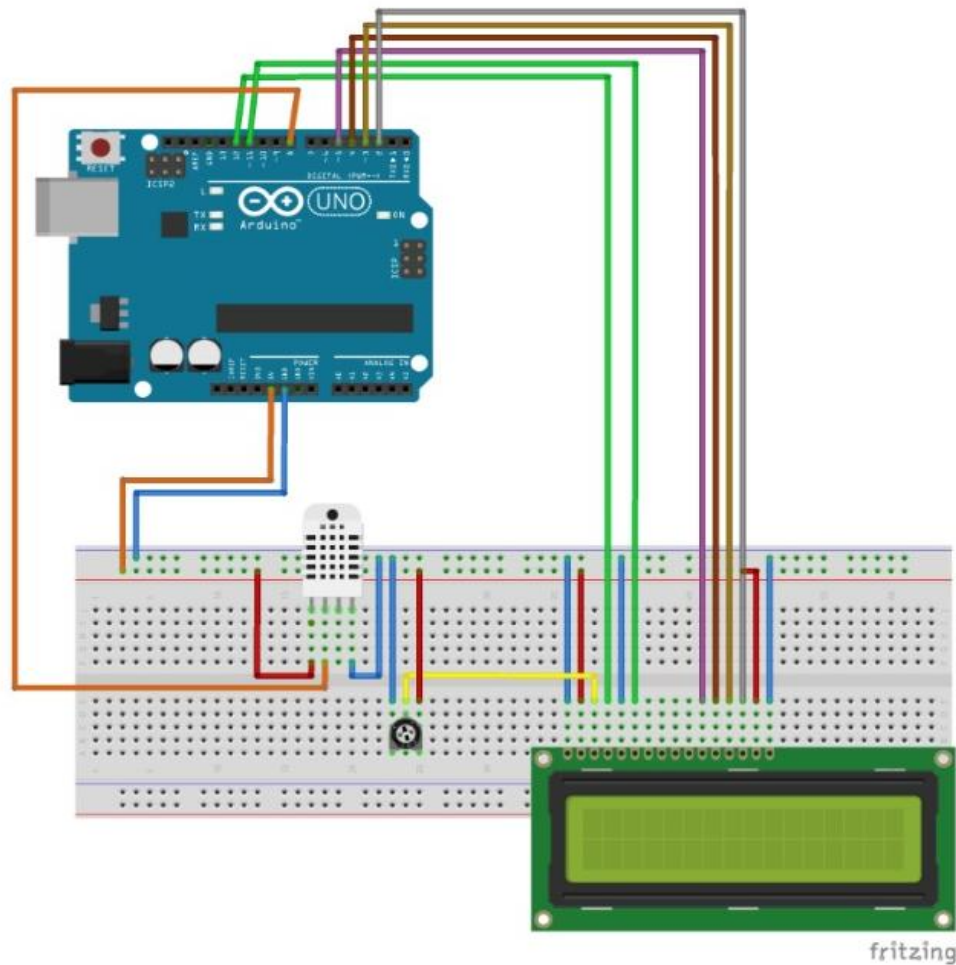
Rear view with
cover removed

The background features several overlapping geometric shapes, primarily diamonds and triangles, in teal, yellow, and green colors. These shapes are arranged in a way that creates a modern, abstract pattern. The teal shapes are located in the top-left, top-right, and bottom-left areas. The yellow shapes are in the top-right and bottom-left areas. The green shapes are in the top-right and bottom-left areas.

“Operation of Measuring

The DHT11 uses just one signal wire to transmit data to the Arduino. Power comes from separate 5V and ground wires. A 10K Ohm pull-up resistor is needed between the signal line and 5V line to make sure the signal level stays high by default (see the datasheet for more info).

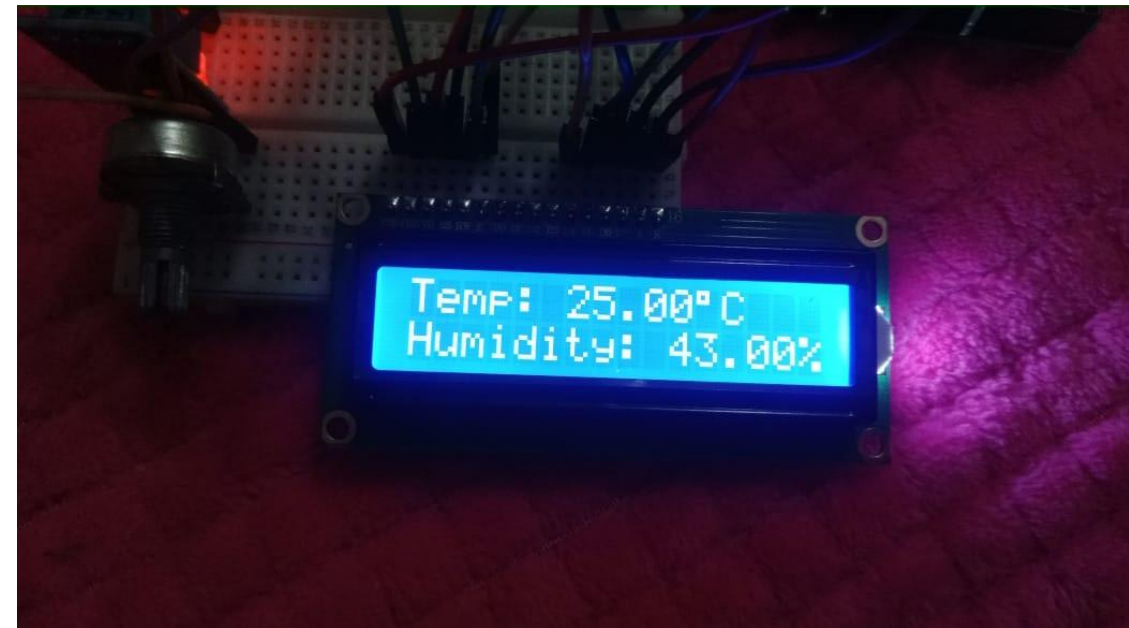
“Connecting with Arduino



“DISPLAY HUMIDITY AND TEMPERATURE ON THE SERIAL MONITOR

you'll need to install the DHTlib library. It has all the functions needed to get the humidity and temperature readings from the sensor.

A nice way to display the humidity and temperature readings is on a 16X4 LCD.



Arduino Code

sketch_dec16a | Arduino 1.8.16

File Edit Sketch Tools Help

```
sketch_dec16a
#include <dht.h>
#include <LiquidCrystal.h>

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

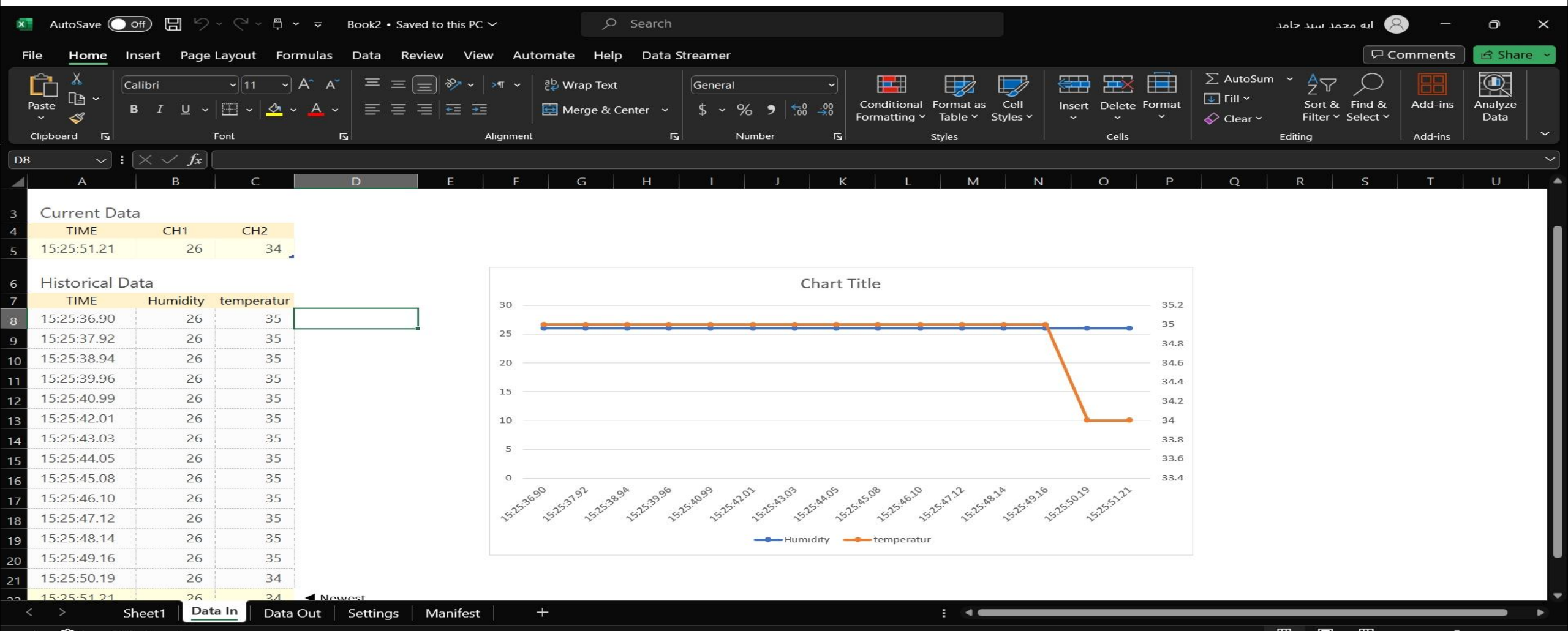
dht DHT;

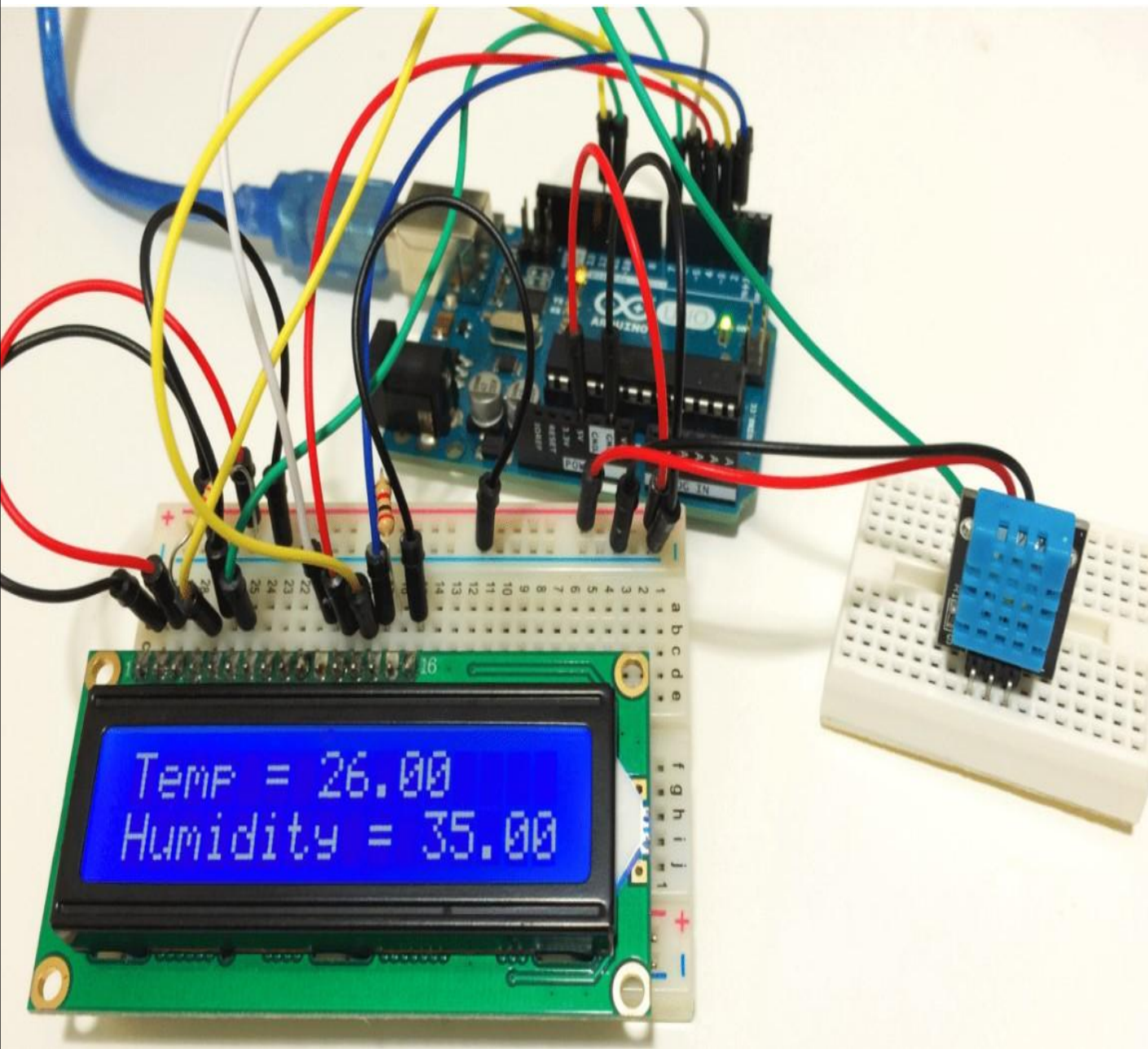
#define DHT11_PIN 7

void setup() {
  lcd.begin(16, 2);
}

void loop() {
  int chk = DHT.read11(DHT11_PIN);
  lcd.setCursor(0,0);
  lcd.print("Temp: ");
  lcd.print(DHT.temperature);
  lcd.print((char)223);
  lcd.print("C");
  lcd.setCursor(0,1);
  lcd.print("Humidity: ");
  lcd.print(DHT.humidity);
  lcd.print("%");
  delay(1000);
}
```

Curve on Excel





Thank you

- HABIBA AHMED FOUAD
- BOSINA SAAD
- AYA MOHAMED